

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5431

5 Credits: 4

Program Core

Source-grounded

Computer Integrated Design

□ To introduce computer integration in all aspects of production system. □
To identify the features of CAM

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5431 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5431.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Mechatronics
Course code	5431
Course title	Computer Integrated Design
Semester	5 Credits: 4
Course category	Program Core
Periods per week	As specified in the official PDF
Periods per semester	As specified in the official PDF
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

12 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES**Objectives, outcomes and mapping****Course objectives**

- □ To introduce computer integration in all aspects of production system. □ To identify the features of CAM
- □ To provide an eye opening to the present scenario of a manufacturing plant so that the
- student can better understand their future job roles.
- □ To familiarize with the automation related to material handling, storage, inspection and manufacturing system

Course outcomes

CO	Official outcome	Study level
CO1	Familiarize the computer integration in design	See official syllabus
CO2	14 Understanding phase of product development. Identify the features of CAM 14 Understanding	See official syllabus
CO3	Describe the automation related to material handling,	See official syllabus
CO4	12 Understanding storage, inspection and manufacturing system.	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3

Row	Extracted official mapping
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Outline the automation opportunities and strategies.	3	As specified
Module 2	development.	2	As specified
Module 3	Identify the features of CAM.	4	As specified
Module 4	and manufacturing system.	3	As specified

MODULE 1

Outline the automation opportunities and strategies.

This module is presented from the official syllabus scope for Computer Integrated Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Outline the automation opportunities and strategies.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ 2 Understanding in manufacturing system. Discuss about automation, its elements and

2 Understanding in manufacturing system. Discuss about automation, its elements and is an official topic in Module 1 of Computer Integrated Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding in manufacturing system. Discuss about automation, its elements and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding in manufacturing system. discuss about automation, its elements and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 2 Understanding in manufacturing system. Discuss about automation, its elements and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-2 Understanding in manufacturing system. Discuss about automation, its elements and definition/meaning. Steps, application. Technical terms English- Malayalam.

▼ 12 Understanding types Discuss the principles and strategies for

12 Understanding types Discuss the principles and strategies for is an official topic in Module 1 of Computer Integrated Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 12 Understanding types Discuss the principles and strategies for using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 12 understanding types discuss the principles and strategies for should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 12 Understanding types Discuss the principles and strategies for means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-12 Understanding types Discuss the principles and strategies for definition/meaning. steps, application. Technical terms English-.

▼ 6 Understanding automation Concept of Computer Integrated Manufacturing (CIM); Opportunities for Automation and Computerization in a Production System- Facilities and Manufacturing Support Systems; Basic elements of automation Sensors - Desirable features, common types of sensors used in automation; Actuators - common types of actuators used in automation; Levels of automation; Types of Automation - Programmable Automation, Flexible Automation and Fixed Automation; Reasons for Automation; USA principle, Ten Strategies for Automation and Process Improvement; Automation Migration Strategy - Manual Production, Automated Production, Automated Integrated Production. Recognize the computer integration in design phase of product

6 Understanding automation Concept of Computer Integrated Manufacturing (CIM); Opportunities for Automation and Computerization in a Production System- Facilities and Manufacturing Support Systems; Basic elements of automation Sensors - Desirable features, common types of sensors used in automation; Actuators - common types of actuators used in automation; Levels of automation; Types of Automation - Programmable Automation, Flexible Automation and Fixed Automation; Reasons for Automation; USA principle, Ten Strategies for Automation and Process Improvement; Automation Migration Strategy - Manual Production, Automated Production, Automated Integrated Production. Recognize the computer integration in design phase of product is an official topic in Module 1 of Computer Integrated Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding automation Concept of Computer Integrated Manufacturing (CIM); Opportunities for Automation and Computerization in a Production System- Facilities and Manufacturing Support Systems; Basic elements of automation Sensors - Desirable features, common types of sensors used in automation; Actuators - common types of actuators used in automation; Levels of automation; Types of Automation - Programmable Automation, Flexible Automation and Fixed Automation; Reasons for Automation; USA principle, Ten Strategies for Automation and Process Improvement; Automation Migration Strategy - Manual Production, Automated Production, Automated Integrated Production. Recognize the computer integration in design phase of product using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding automation concept of computer integrated manufacturing (cim); opportunities for automation and computerization in a production system- facilities and manufacturing support systems; basic elements of automation sensors - desirable features, common types of sensors used in automation; actuators - common types of actuators used in automation; levels of automation; types of automation - programmable automation, flexible automation and fixed automation; reasons for automation; usa principle, ten strategies for automation and process improvement; automation migration strategy - manual production, automated production, automated integrated production. recognize the computer integration in design phase of product should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding automation Concept of Computer Integrated Manufacturing (CIM); Opportunities for Automation and Computerization in a Production System- Facilities and Manufacturing Support Systems; Basic elements of automation Sensors - Desirable features, common types of sensors used in automation; Actuators - common types of actuators used in automation; Levels of automation; Types of Automation - Programmable Automation, Flexible Automation and Fixed Automation; Reasons for Automation; USA principle, Ten Strategies for Automation and Process Improvement; Automation Migration Strategy - Manual Production, Automated Production, Automated Integrated Production. Recognize the computer integration in design phase of product means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 6 Understanding automation Concept of Computer Integrated Manufacturing (CIM); Opportunities for Automation and Computerization in a Production System- Facilities and Manufacturing Support Systems; Basic elements of automation Sensors - Desirable features, common types of sensors used in automation; Actuators - common types of actuators used in automation; Levels of automation; Types of Automation - Programmable Automation, Flexible Automation and Fixed Automation; Reasons for Automation; USA principle, Ten Strategies for Automation and Process Improvement; Automation Migration Strategy - Manual Production, Automated Production, Automated Integrated Production. Recognize the computer integration in design phase of product

രണ്ടാം വർഷത്തിൽ നിന്നും definition/meaning നൽകണം. രണ്ടാം വർഷത്തിൽ
രണ്ടാം വർഷം, steps, application നൽകണം രണ്ടാം വർഷം. Technical terms
English-ൽ രണ്ടാം വർഷം നൽകണം.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2**development.**

This module is presented from the official syllabus scope for Computer Integrated Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: development.
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

▼ Describe about product development. 5 Understanding Describe the computer-aided designing and

Describe about product development. 5 Understanding Describe the computer-aided designing and is an official topic in Module 2 of Computer Integrated Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe about product development. 5 Understanding Describe the computer-aided designing and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe about product development. 5 understanding describe the computer-aided designing and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe about product development. 5 Understanding Describe the computer-aided designing and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Describe about product development. 5 Understanding Describe the computer-aided designing and definition/meaning . steps, application . Technical terms English- .

▼ 9 Understanding prototyping. Product Development - Product Development Cycle and Product Development Spiral. Sequential and Concurrent Engineering (Introduction Level Only), Comparison. Computer Aided Design (CAD): Definition, Advantages, CAD hardware (Name and function) and software (Name Only); Geometric Modelling - Wireframe, Surface and Solid Modelling. Rapid Prototyping - Liquid Based, Solid Based and Powder Based Prototyping

9 Understanding prototyping. Product Development - Product Development Cycle and Product Development Spiral. Sequential and Concurrent Engineering (Introduction Level Only), Comparison. Computer Aided Design (CAD): Definition, Advantages, CAD hardware (Name and function) and software (Name Only); Geometric Modelling - Wireframe, Surface and Solid Modelling. Rapid Prototyping - Liquid Based, Solid Based and Powder Based Prototyping is an official topic in Module 2 of Computer Integrated Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 9 Understanding prototyping. Product Development - Product Development Cycle and Product Development Spiral. Sequential and Concurrent Engineering (Introduction Level Only), Comparison. Computer Aided Design (CAD): Definition, Advantages, CAD hardware (Name and function) and software (Name Only); Geometric Modelling - Wireframe, Surface and Solid Modelling. Rapid Prototyping - Liquid Based, Solid Based and Powder Based Prototyping using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 9 understanding prototyping. product development – product development cycle and product development spiral. sequential and concurrent engineering (introduction level only), comparison. computer aided design (cad): definition, advantages, cad hardware (name and function) and software (name only); geometric modelling – wireframe, surface and solid modelling. rapid prototyping - liquid based, solid based and powder based prototyping should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 9 Understanding prototyping. Product Development – Product Development Cycle and Product Development Spiral. Sequential and Concurrent Engineering (Introduction Level Only), Comparison. Computer Aided Design (CAD): Definition, Advantages, CAD hardware (Name and function) and software (Name Only); Geometric Modelling – Wireframe, Surface and Solid Modelling. Rapid Prototyping - Liquid Based, Solid Based and Powder Based Prototyping means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-9 Understanding prototyping. Product Development – Product Development Cycle and Product Development Spiral. Sequential and Concurrent Engineering (Introduction Level Only), Comparison. Computer Aided Design (CAD): Definition, Advantages, CAD hardware (Name and function) and software (Name Only); Geometric Modelling – Wireframe, Surface and Solid Modelling. Rapid Prototyping - Liquid Based, Solid Based and Powder Based Prototyping ഏകദേശം 1000 വാക്ക് definition/meaning 1000 വാക്ക്. 1000 വാക്ക് 1000 വാക്ക്, steps, application 1000 വാക്ക് 1000 വാക്ക് 1000 വാക്ക്. Technical terms English-1000 വാക്ക് 1000 വാക്ക്.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Identify the features of CAM.

This module is presented from the official syllabus scope for Computer Integrated Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Identify the features of CAM.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Describe CAM, CAPP 3 Remembering Explain the Variant and Generative techniques

Describe CAM, CAPP 3 Remembering Explain the Variant and Generative techniques is an official topic in Module 3 of Computer Integrated Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe CAM, CAPP 3 Remembering Explain the Variant and Generative techniques using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe cam, capp 3 remembering explain the variant and generative techniques should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Describe CAM, CAPP 3 Remembering Explain the Variant and Generative techniques means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Describe CAM, CAPP 3 Remembering Explain the Variant and Generative techniques definition/meaning . steps, application . Technical terms English- .

▼ 4 Understanding in CAPP Describe MPS, MRP, CRP and inventory

4 Understanding in CAPP Describe MPS, MRP, CRP and inventory is an official topic in Module 3 of Computer Integrated Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding in CAPP Describe MPS, MRP, CRP and inventory using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding in capp describe mps, mrp, crp and inventory should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 4 Understanding in CAPP Describe MPS, MRP, CRP and inventory means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 4 Understanding in CAPP Describe MPS, MRP, CRP and inventory നോക്കൂ definition/meaning നോക്കൂ. നോക്കൂ നോക്കൂ, steps, application നോക്കൂ നോക്കൂ. Technical terms English- നോക്കൂ നോക്കൂ.

▼ 3 Remembering control with aid of computer.

3 Remembering control with aid of computer. is an official topic in Module 3 of Computer Integrated Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Remembering control with aid of computer. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 remembering control with aid of computer. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Remembering control with aid of computer. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 3 Remembering control with aid of computer. ഫലപ്രദമായി പ്രകടിപ്പിക്കുക definition/meaning പ്രകടിപ്പിക്കുക. പ്രകടിപ്പിക്കുക പ്രകടിപ്പിക്കുക, steps, application പ്രകടിപ്പിക്കുക പ്രകടിപ്പിക്കുക. Technical terms English- പ്രകടിപ്പിക്കുക പ്രകടിപ്പിക്കുക.

▼ Discuss about CNC Machines 4 Understanding Computer Aided Manufacturing (CAM)- Definition; Computer Aided Process Planning (CAPP)- Definition; Group Technology (Introduction Level treatment of cellular manufacturing, part family and its methods), Variant and Generative CAPP. Master Production Schedule (MPS), Material Requirements Planning (MRP), Capacity Requirement Planning (CRP) and Inventory Control with the aid of Computer (Introduction Level only). CNC and DNC- Features and Advantages; Machining Centres - Turning centre, Milling Centre. Discuss the automation related to material handling, storage, inspection

Discuss about CNC Machines 4 Understanding Computer Aided Manufacturing (CAM)- Definition; Computer Aided Process Planning (CAPP)- Definition; Group Technology (Introduction Level treatment of cellular manufacturing, part family and its methods), Variant and Generative CAPP. Master Production Schedule (MPS), Material Requirements Planning (MRP), Capacity Requirement Planning (CRP) and Inventory Control with the aid of Computer (Introduction Level only). CNC and DNC- Features and Advantages; Machining Centres – Turning centre, Milling Centre. Discuss the automation related to material handling, storage, inspection is an official topic in Module 3 of Computer Integrated Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss about CNC Machines 4 Understanding Computer Aided Manufacturing (CAM)- Definition; Computer Aided Process Planning

(CAPP)- Definition; Group Technology (Introduction Level treatment of cellular manufacturing, part family and its methods), Variant and Generative CAPP. Master Production Schedule (MPS), Material Requirements Planning (MRP), Capacity Requirement Planning (CRP) and Inventory Control with the aid of Computer (Introduction Level only). CNC and DNC- Features and Advantages; Machining Centres – Turning centre, Milling Centre. Discuss the automation related to material handling, storage, inspection using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss about cnc machines 4 understanding computer aided manufacturing (cam)- definition; computer aided process planning (capp)- definition; group technology (introduction level treatment of cellular manufacturing, part family and its methods), variant and generative capp. master production schedule (mps), material requirements planning (mrp), capacity requirement planning (crp) and inventory control with the aid of computer (introduction level only). cnc and dnc- features and advantages; machining centres – turning centre, milling centre. discuss the automation related to material handling, storage, inspection should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss about CNC Machines 4 Understanding Computer Aided Manufacturing (CAM)- Definition; Computer Aided Process Planning (CAPP)- Definition; Group Technology (Introduction Level treatment of cellular manufacturing, part family and its methods), Variant and Generative CAPP. Master Production Schedule (MPS), Material Requirements Planning (MRP), Capacity Requirement Planning (CRP) and Inventory Control with the aid of Computer (Introduction Level only).

Aspect	What to prepare
	CNC and DNC- Features and Advantages; Machining Centres – Turning centre, Milling Centre. Discuss the automation related to material handling, storage, inspection means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Discuss about CNC Machines 4 Understanding Computer Aided Manufacturing (CAM)- Definition; Computer Aided Process Planning (CAPP)- Definition; Group Technology (Introduction Level treatment of cellular manufacturing, part family and its methods), Variant and Generative CAPP. Master Production Schedule (MPS), Material Requirements Planning (MRP), Capacity Requirement Planning (CRP) and Inventory Control with the aid of Computer (Introduction Level only). CNC and DNC- Features and Advantages; Machining Centres – Turning centre, Milling Centre. Discuss the automation related to material handling, storage, inspection. definition/meaning. steps, application. Technical terms English-.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4**and manufacturing system.**

This module is presented from the official syllabus scope for Computer Integrated Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: and manufacturing system.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ 4 Understanding and storage systems Describe various automated inspection

4 Understanding and storage systems Describe various automated inspection is an official topic in Module 4 of Computer Integrated Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding and storage systems Describe various automated inspection using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding and storage systems describe various automated inspection should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What 4 Understanding and storage systems Describe various automated inspection means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 4 Understanding and storage systems Describe various automated inspection definition/meaning . , steps, application . Technical terms English- .

▼ 4 Understanding technologies. Describe the manufacturing systems and factors

4 Understanding technologies. Describe the manufacturing systems and factors is an official topic in Module 4 of Computer Integrated Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding technologies. Describe the manufacturing systems and factors using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding technologies. describe the manufacturing systems and factors should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 4 Understanding technologies. Describe the manufacturing systems and factors means in the official course context
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Malayalam support: Module 4- 4 Understanding technologies. Describe the manufacturing systems and factors definition/meaning . steps, application . Technical terms English- .

▼ 4 Understanding for selection. Material handling - Fixed Routing and Variable Routing; List of Material Handling Equipments for Each Routing. Storage - Automated Storage/Retrieval System (AS/RS) and Carousel Storage System (Introduction Level Only). Inspection Technologies - Comparison of Contact and Non-Contact Inspection Techniques, CMM and Machine Vision (Introduction with Simple Diagram). Manufacturing Systems - Factors for Selection, Types - Single Station Cell, Multistation Fixed Routing, Multistation variable Routing (Introduction Level Only]. Text / Reference T/R Book Title/Author T1 Mikell P. Groover, Automation, Production Systems and Computer-Integrated Manufacturing. 4th edn. Pearson. T2 P.Radhakrishnan S.Subramanyan and V. Raju, CAD/CAM/CIM. 4th edn. New Age International Publishers. R1 S.R. Deb and S. Deb, Robotics Technology and Flexible Automation, Tata McGraw Hill. Online resources S.No Website Link 1 <https://nptel.ac.in/courses/112/104/112104289/> 2 <https://www.sciencedirect.com/search?qs=computer%20integrated%20manufacturing>

4 Understanding for selection. Material handling - Fixed Routing and Variable Routing; List of Material Handling Equipments for Each Routing. Storage - Automated Storage/Retrieval System (AS/RS) and Carousel Storage System (Introduction Level Only). Inspection Technologies - Comparison of Contact and Non-Contact Inspection Techniques, CMM and Machine Vision (Introduction with Simple Diagram). Manufacturing Systems - Factors for Selection, Types - Single Station Cell, Multistation Fixed Routing, Multistation variable Routing (Introduction Level Only]. Text / Reference T/R Book Title/Author T1 Mikell P. Groover, Automation, Production Systems and Computer-Integrated Manufacturing. 4th edn. Pearson. T2 P.Radhakrishnan S.Subramanyan and V. Raju, CAD/CAM/CIM. 4th edn. New Age International Publishers. R1 S.R. Deb and S. Deb, Robotics Technology and Flexible Automation, Tata McGraw Hill. Online resources S.No Website Link 1 <https://nptel.ac.in/courses/112/104/112104289/> 2 [https://www.sciencedirect.com/search?](https://www.sciencedirect.com/search?qs=computer%20integrated%20manufacturing)

qs=computer%20integrated%20manufactur ing is an official topic in Module 4 of Computer Integrated Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

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To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

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Aspect	What to prepare
Definition/identity	What 4 Understanding for selection. Material handling – Fixed Routing and Variable Routing; List of Material Handling Equipments for Each Routing. Storage – Automated Storage/Retrieval System (AS/RS) and Carousel Storage System (Introduction Level Only). Inspection Technologies – Comparison of Contact and Non-Contact Inspection Techniques, CMM and Machine Vision (Introduction with Simple Diagram). Manufacturing Systems – Factors for Selection, Types – Single Station Cell, Multistation Fixed Routing, Multistation variable Routing (Introduction Level Only]. Text / Reference T/R Book Title/Author T1 Mikell P. Groover, Automation, Production Systems and Computer-Integrated Manufacturing. 4th edn. Pearson. T2 P.Radhakrishnan S.Subramanyan and V. Raju, CAD/CAM/CIM. 4th edn. New Age International Publishers. R1 S.R. Deb and S. Deb, Robotics Technology and Flexible Automation, Tata McGraw Hill. Online resources S.No Website Link 1 https://nptel.ac.in/courses/112/104/112104289/ 2 https://www.sciencedirect.com/search?qs=computer%20integrated%20manufacturing

Aspect	What to prepare
	qs=computer%20integrated%20manufactur ing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 4 Understanding for selection. Material handling – Fixed Routing and Variable Routing; List of Material Handling Equipments for Each Routing. Storage – Automated Storage/Retrieval System (AS/RS) and Carousel Storage System (Introduction Level Only). Inspection Technologies – Comparison of Contact and Non-Contact Inspection Techniques, CMM and Machine Vision (Introduction with Simple Diagram). Manufacturing Systems – Factors for Selection, Types – Single Station Cell, Multistation Fixed Routing, Multistation variable Routing (Introduction Level Only]. Text / Reference T/R Book Title/Author T1 Mikell P. Groover, Automation, Production Systems and Computer-Integrated Manufacturing. 4th edn. Pearson. T2 P.Radhakrishnan S.Subramanyan and V. Raju, CAD/CAM/CIM. 4th edn. New Age International Publishers. R1 S.R. Deb and S. Deb, Robotics Technology and Flexible Automation, Tata McGraw Hill. Online resources S.No Website Link 1 <https://nptel.ac.in/courses/112/104/112104289/> 2 <https://www.sciencedirect.com/search?qs=computer%20integrated%20manufactur ing> □□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□. □□□□□ □□□□□ □□□□□□, steps,

application ഏകദേശം അളക്കുന്നതിനായി ഉപയോഗിക്കുന്നു. Technical terms English-ൽ ഉപയോഗിക്കുന്നു.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► **Define or explain 2 Understanding in manufacturing system. Discuss about automation, its elements and. (3 marks)**

► **Define or explain 12 Understanding types Discuss the principles and strategies for. (3 marks)**

► **Define or explain 6 Understanding automation Concept of Computer Integrated Manufacturing (CIM); Opportunities for Automation and Computerization in a Production System- Facilities and Manufacturing Support Systems; Basic elements of automation Sensors - Desirable features, common types of sensors used in automation; Actuators - common types of actuators used in automation; Levels of automation; Types of Automation - Programmable Automation, Flexible Automation and Fixed Automation; Reasons for Automation; USA principle, Ten Strategies for Automation and Process Improvement; Automation Migration Strategy - Manual Production, Automated Production, Automated Integrated Production. Recognize the computer integration in design phase of product. (3 marks)**

Module 2

► **Define or explain Describe about product development. 5 Understanding Describe the computer-aided designing and. (3 marks)**

► **Define or explain 9 Understanding prototyping. Product Development - Product Development Cycle and Product Development Spiral. Sequential and Concurrent Engineering (Introduction Level Only), Comparison. Computer Aided Design (CAD): Definition, Advantages, CAD hardware (Name and function) and software (Name Only); Geometric Modelling - Wireframe, Surface and Solid Modelling. Rapid Prototyping - Liquid Based, Solid Based and Powder Based Prototyping. (3 marks)**

Module 3

► **Define or explain Describe CAM, CAPP 3 Remembering Explain the Variant and Generative techniques. (3 marks)**

► **Define or explain 4 Understanding in CAPP Describe MPS, MRP, CRP and inventory. (3 marks)**

► **Define or explain 3 Remembering control with aid of computer.. (3 marks)**

► **Define or explain Discuss about CNC Machines 4 Understanding Computer Aided Manufacturing (CAM)- Definition; Computer Aided Process Planning (CAPP)- Definition; Group Technology (Introduction Level treatment of cellular manufacturing, part family and its methods), Variant and Generative CAPP. Master Production Schedule (MPS), Material Requirements Planning (MRP), Capacity Requirement Planning (CRP) and Inventory Control with the aid of Computer (Introduction Level only). CNC and DNC- Features and Advantages; Machining Centres - Turning centre, Milling Centre. Discuss the automation related to material handling, storage, inspection. (3 marks)**

Module 4

► **Define or explain 4 Understanding and storage systems Describe various automated inspection. (3 marks)**

► **Define or explain 4 Understanding technologies. Describe the manufacturing systems and factors. (3 marks)**

► **Define or explain 4 Understanding for selection. Material handling - Fixed Routing and Variable Routing; List of Material Handling Equipments for Each Routing. Storage - Automated Storage/Retrieval System (AS/RS) and Carousel Storage System (Introduction Level Only). Inspection Technologies - Comparison of Contact and Non-Contact Inspection Techniques, CMM and Machine Vision (Introduction with Simple Diagram). Manufacturing Systems - Factors for Selection, Types - Single Station Cell, Multistation Fixed Routing, Multistation variable Routing (Introduction Level Only]. Text / Reference T/R Book**
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P.Radhakrishnan S.Subramanyan and V. Raju, CAD/CAM/CIM. 4th edn. New Age International Publishers. R1 S.R. Deb and S. Deb, Robotics Technology and Flexible Automation, Tata McGraw Hill. Online resources
S.No Website Link 1 <https://nptel.ac.in/courses/112/104/112104289/> 2
<https://www.sciencedirect.com/search?q=computer%20integrated%20manufacturing>. (3 marks)

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **2 Understanding in manufacturing system. Discuss about automation, its elements and.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Describe about product development. 5 Understanding Describe the computer-aided designing and.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Describe CAM, CAPP 3 Remembering Explain the Variant and Generative techniques.**

Module 4

1-mark definition: State the meaning of **4 Understanding and storage systems Describe various automated inspection.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- S.No Website Link
- <https://nptel.ac.in/courses/112/104/112104289/>
- <https://www.sciencedirect.com/search?qs=computer%20integrated%20manufactur>

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 5431. Verify institutional updates against the current official curriculum.